

CS383

ميد هندسة برمجيات محلول
ترم 461

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التحديد :
لكتشر ١ الى ٧

ادعوا لامي بالشفاء والصحه والعافيه
وبالتوفيق لكم يارب



Course Code: CS 383: Software Engineering
MID Exam
461 - Date: 10/04/2024 Duration: 1 hour

Student Name: _____

Student ID: _____

CLOs	k1	k2	k3	S1	S2	S3	S4	V1	V2
Question number	1,3					4,5			2
Allocated marks	6					7			2
Earned marks									

Total Marks

15 /15

The exam starts here

Q1. Choose the correct answer, then full up the table with correct choice

L = Lecture
S = Slide

- What does product quality depend on?
 a. Development process
 b. Developer skills
 c. Project size
 d. All of the above
- What is one of the process attributes that focuses on how easy it is to understand the process definition?
 a. Standardization
 b. Understandability
 c. Visibility
 d. Measurability
- Project schedule chart include:
 a. tasks
 b. people involved for each task
 c. milestones
 d. all of them
- Project manager should consider four factors in people management include:
 a. Consistence, Respect, Loyalty, Honesty
 b. Consistence, Respect, Speed, Experience
 c. Consistence, Experience, Inclusion, Honesty
 d. Consistence, Respect, Inclusion, Honesty
- Risk management involves anticipating risks that may affect:
 a. the project schedule
 b. the quality of project
 c. a and b
 d. none of them
- Which of the following is not considered as a risk in project management?
 a. Specification delays
 b. Product competition
 c. Testing
 d. Hardware delays

1	2	3	4	5	6

L.2

Q2. According to the 8 principles of the IEEE Code of Ethics (PUBLIC, CLIENT AND EMPLOYER, PRODUCT, JUDGMENT, MANAGEMENT, PROFESSION, COLLEAGUES, SELF), identify the type of ethics reflected in the following descriptions:

Points/2

SEEPP ethic example	Ethics Principle
Approve software only if they believe that it is safe for use, meets specifications, passes appropriate tests	Public
Ensure proper and achievable goals and objectives for any project	Product
Assist colleagues in being fully aware of current standard work practices	colleagues
Improve their ability to produce accurate, informative, and well-written documentation	Self

Q3. For each concern, determine whether the Plan Driven or Agile method is the most suitable choice:

Points/3

Concern	Plan Driven	Agile
1- Emphasis on cost estimation before starting	✓	
2- Flexibility in handling changing requirements		✓
3- Use of detailed risk management	✓	
4- Incremental delivery of features		✓
5- Long periods of planning and analysis	✓	
6- Continuous integration and testing		✓

L.3 5.8

Q4. What are the 4 key software engineering activities. The activities must be given in a logical order to ensure the successful delivery of software products:

Points/2

1) **Process activities of software development**

2) In the last, we mentioned fundamental software development activities:

- 3) .
- 4) ..
1. Software specification
 2. Software development
 3. Software verification and validation
 4. Software evolution

Q5. After reading the following system description determine the functional and non-functional requirements:

Points/5

L.7

A company hires out tools and equipment (for example drills, power saws, cement mixers, ladders, scaffolding) to customers and requires a computerized system to record details of bookings. Equipment may be booked in advance, or customers may appear at the reception desk and ask if there is an item available for immediate hire. When dealing with a booking or allocating an available item to a customer, the receptionist must check whether the customer has previously hired equipment from the company or is a new customer. For a new customer, the receptionist must enter the customer's details. Otherwise, the receptionist must retrieve the existing customer's record and update any details if necessary. Each page must load within 5 seconds. The minimum period of hire is one day, and all hires are made for several complete days. The return of an item at the end of the hire period is recorded by the receptionist, or by a technician if the reception desk is closed. The system should be available at working hours. The manager of the company requires a summary of the status of all equipment at the beginning of each day, giving details of items out on hire, items booked and items that will be available for hire that day.

Functional Requirements

Retrieve customer's details

Update customer's details

Add new customer

Check availability of tools

Make an advance booking

Record an immediate hire

Book out a tool

Record return of a tool

Produce management summary

Nonfunctional Requirements

Non-functional requirements

The fact that a hire is for a whole number of days could be seen as a non-functional requirement; alternatively this could lead to a new functional requirement to allow the minimum hire period to be specified. But probably the restriction to whole numbers of days would permeate the system when we get on to more detailed design, so it makes most sense as a non-functional requirement.